www.Padasalai.Net www.CBSEtips.in

QUARTERLY EXAMINATION - 2023

CHEMISTRY

CLASS : XII

Register No

TIME : 3.00 Hr.

MAX. MARKS : 70

PART - I

I) Choose the Correct Answer. 15 X 1 = 15

1. Considering Ellingham diagram, which of the following metals can be used to reduce alumina?
   a) Fe b) Cu c) Mg d) zn
2. Match the items in column - I with the items of column- II and assign the correct code

Column - I	Column- II		A	B	C	D
A) Cyanide process -	1. Ultrapure Ge	(a)	2	3	1	4
B) Froth floatation process-	2. Dressing of Zns	(b)	4	2	1	3
C) Electrolytic reduction-	3. Extraction of Au	(c)	1	2	3	4
D) Zone refining -	4. Extraction of Al	(d)	3	4	1	2

3. The geometry at which Carbon atom in diamond are bonded to each other is
   a) Teterahedral b) hexagonal c) Octahedral d) none of these
4. The formula for Hyponitrous acid is
   a) HOONO b) HNO_2 c) $H_2N_2O_2$ d) HNO_4
5. The most common oxidation state of actinoids is
   a) +2 b) +3 c) +4 d) +6
6. Assertion : due to Frenkel defect, density of the crystalline solid decreases
   Reason : In Frenkel defect cation and anion leaves the crystal.
   a) Both assertion and reason are true and reason is correct explanation of assertion
   b) Both assertion and reason are true but reason is not the correct explanation of assertion
   c) Assertion is true but reason is false d) Both assertion and reason are false
7. The crystal with a metal deficiencey defect is
   a) Nacl b) Feo c) Zno d) Kcl
8. If 75% of a first order reaction was completed in 40 minutes, 50 % of the same reaction under the same conditions would be completed in
   a) 20 minutes b) 30 minutes c) 35 minutes d) 75 minutes
9. The addition of a catalyst during a chemical reaction alters which of the following quantities?
   a) Enthalpy b) Activation energy c) Internal energy d) Entropy
10. The P^H of an aqueous solution is Zero the solution is
    a) neutral b) slightyly acidic c) Strongly acidic d) basic
11. If the solubility product of Lead iodide is 3.2×10^{-8}, its solubility will be
    a) $2X10^{-3}$ M b) 4×10^{-4} M c) 1.6×10^{-5} M d) 1.8×10^{-5} M
12. In Victor mayer test the secondary alcohol gives the colour is
    a) Red b) Yellow c) Green d) Blue
13. Which of the following compound can be used as antifreeze in automobile radiators?
    a) Ethylene glycol b) Methanol c) Ethanol d) Phenol
14. Which one of the following reaction is an example of disproportionation reaction?
    a) Aldol condensation b) Cannizaro reaction
    c) Benzoin Condensation d) None of these
15. Which one of the following reducees tollens reagent
    a) Formic acid b) Acetic acid c) Benzophenone d) None of these

Kindly send me your study materials to padasalai.net@gmail.com

PART - II

II) Answer any 6 Questions (Q.No.24 is Compulsory) **6X2=12**

16. Which type of ores can be concentranted by froth floatation method? Give two example for such ores.
17. Write the anamolous Property of the first element of 'P' - block.
18. Why fluorine is more reactive than other halogens?
19. Define packing efficiency of crystals.
20. What is half life of a reaction ?
21. Identify the conjugate acid - base pair of the following reaction

    (i) $HS^-_{(aq)} + HF \rightleftharpoons F^-_{(aq)} + H_2S_{(aq)}$ (ii) $HPO_4^{2-} + SO_3^{2-} \rightleftharpoons PO_4^{3-} + HSO_3^-$
22. Give short note on Dows Process.
23. What is Urotrophine? Write the uses of urotrophine.
24. Which is more stable Fe^{3+}(or) Fe^{3+} ? Explain.

PART - III

III) Answer any 6 Questions (Q.No.33 is Compulsory) **6X3=18**

25. What are the difference between minerals and Ore.
26. Write the uses of Helium.
27. Complete the equations. (i) $HCOOH + H_2SO_4 \longrightarrow$? (ii) $H_2B_4O_7 \xrightarrow{\text{red hot}}$?
28. Why transition elements forms complex compounds?
29. Write the difference between metal excess defect and metal deficiency defect.
30. Define rate law and rate constant.
31. Write the Rosenmund reduction reaction?
32. Write the dehydrating reaction of Glycerol
33. Write the expression for the solubility product of $Ca_3(PO_4)_2$.

PART - IV

IV) Answer all the questions : **5X5=25**

34. a) Explain Froth floatation method with diagram. (5)

    (OR)

    b) i) Write the Oxidation state of halogens for the following compounds. 1) OF_2 2) I_2O_4 (2)

    ii) Write the short note on bleaching action of chlorine. (3)

35. a) i) Explain Structure of diborane. (3)

    ii) Write a short note on Fisher tropsch Synthesis. (2)

    (OR)

    b) Compare Lanthanoids and Actinoids. (5)

36. a) i) To Calculate the number of atoms present in sc,bcc, fcc unit cell (3)

    ii) Define unit cell. Write the seven type of unit cell. (2)

    (OR)

    b) Derive integrated rate equations for Zero order reaction. (5)

37. a) i) Write Arrhenius equation and explain the terms involved . (2)

    ii) The rate constant for a first order reaction is $1.54 \times 10^{-3}\ S^{-1}$ Calculate its half life time. (3)

    (OR)

    b) Derive the expression for Ostwald's dilution law. (5)

38. a) i) Write the coupling reaction of phenol (2)

    ii) Explain three methods of preparation of ether. (3)

    (OR)

    b) Write the mechanism of Aldol condensation. (5)

****

Kindly send me your study materials to padasalai.net@gmail.com

www.Padasalai.Net www.CBSEtips.in

QUARTERLY EXAMINATION - 2023

CHEMISTRY

CLASS : XII MAX MARKS : 70

PART-A

1. CHOOSE THE CORRECT ANSWER.

1. c). Mg
2. NO option correct :

A	B	C	D
3	2	4	1

3. a) Tetrahedral
4. c) $H_2N_2O_2$
5. b) +3
6. d) Both assertion and reason are false.
7. b) FeO
8. a) 20 minutes
9. b) Activation energy
10. c) Strongly acidic
11. a) 2×10^{-3} M
12. d) Blue
13. a) Ethylene glycol
14. b) Cannizaro reaction
15. a) Formic acid.

பகுதி-அ

I. சரியான விடையைத் தேர்ந்தெடுத்து எழுதுக.

1. இ) mg
2.

A	B	C	D
3	2	4	1

3. அ) நான்முகி
4. இ) $H_2N_2O_2$
5. அ) +3
6. ஈ) கூற்று மற்றும் காரணம் இரண்டும் தவறு
7. அ) FeO
8. அ) 20 நிமிடங்கள்
9. அ) வினை ஆற்றல்
10. இ) அதிக அமிலத்தன்மை கொண்டது.
11. அ) 2×10^{-3} M
12. ஈ) நீலம்
13. இ) எத்திலீன் கிளைகால்
14. அ) கானிசாரோ வினை
15. அ) ∴ பார்மிக் அமிலம்

Answer : 8.

$t_{75\%} = 2\, t_{50\%}$.

$t_{50\%} = \frac{t_{75\%}}{2}$

$t_{50\%} = \frac{40}{2} = 20$

13. IUPAC NAME :
ethan-1,2-diol
Common NAME : Ethylene glycol.

Kindly send me your study materials to padasalai.net@gmail.com

www.Padasalai.Net www.CBSEtips.in

PART-II

11. [Q.NO 24 is compulsory]

24. Which is more stable Fe^{3+} (or) Fe^{2+}? Explain

Fe^{3+} ion is more stable due to its half filled $3d^5$ configuration. but Fe^{2+} ion has $3d^6$ configuration.

PART-III

[Q.NO 33 is compulsory]

33. Write the expression for the solubility product of $Ca_3(PO_4)_2$.

Soln:

$$Ca_3(PO_4)_2 \rightleftharpoons 3Ca^{2+} + 2PO_4^{3-}$$

$$S \qquad 3S \qquad 2S$$

$$K_{SP} = (3S)^3 (2S)^2$$

$$K_{SP} = 27S^3 \, 4S^2$$

$$K_{SP} = 108 \, S^5$$

M.Poovarasan M.Sc B.Ed
PG Asst in Chemistry
Dharmapuri District

M.Poovarasan M.Sc B.Ed www.Padasalai.Net

வினா எண். 24 க்கு கட்டாயம் விடையளிக்கவும்.

24. Fe^{3+} மற்றும் Fe^{2+} ல் எது அதிக நிலைப்புத்தன்மை உடையது ஏன்?

* Fe^{3+} வெளிவட்ட எலக்ட்ரான் அமைப்பு $3d^5$.

* Fe^{2+} வெளிவட்ட எலக்ட்ரான் அமைப்பு $3d^6$.

Fe^{3+} சமபாதியாக நிரம்பிய எலக்ட்ரான் அமைப்பை பெற்றுள்ளதால் நிலைப்புத் தன்மை பெறுகிறது.

வினா எண். 33 க்கு கட்டாயம் விடையளிக்கவும்.

33. $Ca_3(PO_4)_2$ ன் கரைதிறன் பெருக்கத்திற்கான சமன்பாட்டை எழுதுக.

$$Ca_3(PO_4)_2 \longrightarrow 3Ca^{2+} + 2PO_4^{3-}$$

$$K_{SP} = [Ca^{2+}]^3 [PO_4^{3-}]^2$$

$$K_{SP} = (3S)^3 (2S)^2$$

$$K_{SP} = 27S^3 \, 4S^2$$

$$K_{SP} = 108 \, S^5$$

Kindly send me your study materials to padasalai.net@gmail.com